Artificial Intelligence

AIML320/420

Tutorial in week 1

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AIML320/420 Week 1

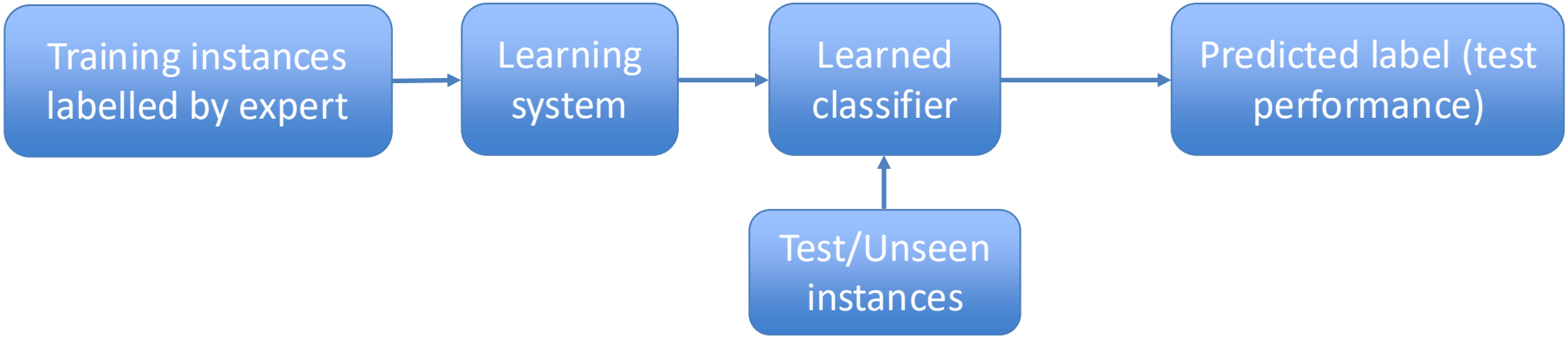
1. Announcements
   - Class rep
   - Helpdesk from week 2: **Tuesday and Thursday 2-3pm at CO242B**
   - In case, programming skill practice is needed, examples are provided, available on course website

Week	Num	Lecturer	Topic	Material
1	1	Fangfang	Course Overview	pptx pdf
^	2	Fangfang	Machine Learning Fundamentals	pptx pdf
^	3	Fangfang	Tutorial 1	pptx pdf ipynb Iris.csv

2. Machine fundamentals
   - Supervised learning and unsupervised learning
   - Regression, classification and clustering
   - Dataset split: training, test and validation set
   - Generalisation and overfitting

A Typical Supervised Learning System

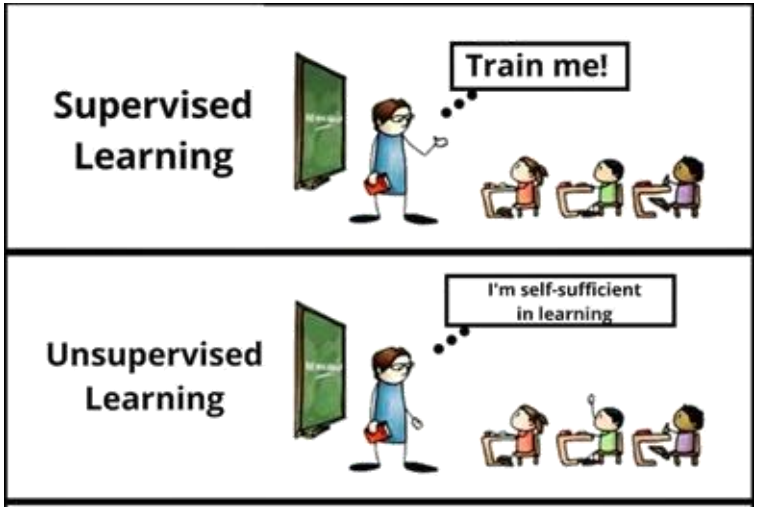
- Presented with a set of training instances
- Need to come up with a model (e.g., rule, pattern) that
  - distinguishes positive examples from negative ones, i.e., classifier
  - find a function for mapping data to targets, i.e., regression model



- **Training set**: a collection of instances from which a classifier is induced/trained
- **Test Set**: A collection of instances which were never used for learning the classifier
  - For measuring the performance of the learnt classifier

(un)Supervised Learning

- Depends on the data are **labelled (supervised) or not (unsupervised)**
- Difference between supervised and unsupervised learning?
  - Input data is labelled VS input data is unlabelled
  - Classification and regression VS clustering
  - Use training dataset VS use just input dataset
  - Used for prediction VS used for analysis



Understanding Data

Features

Labels/ Targets

Instances/ examples

Size	Color	Shape	Taste	Price
L	Red	Square	Good	$15
M	Blue	Circle	Bad	$10
S	Red	Triangle	Good	$7
M	Yellow	Square	Bad	$19

- Number of instances, i.e., need enough instances for training models
- Number of features, i.e., can be small, can be very large (high dimension)
- Features can have different types of values, i.e., need preprocessing

Classification Example

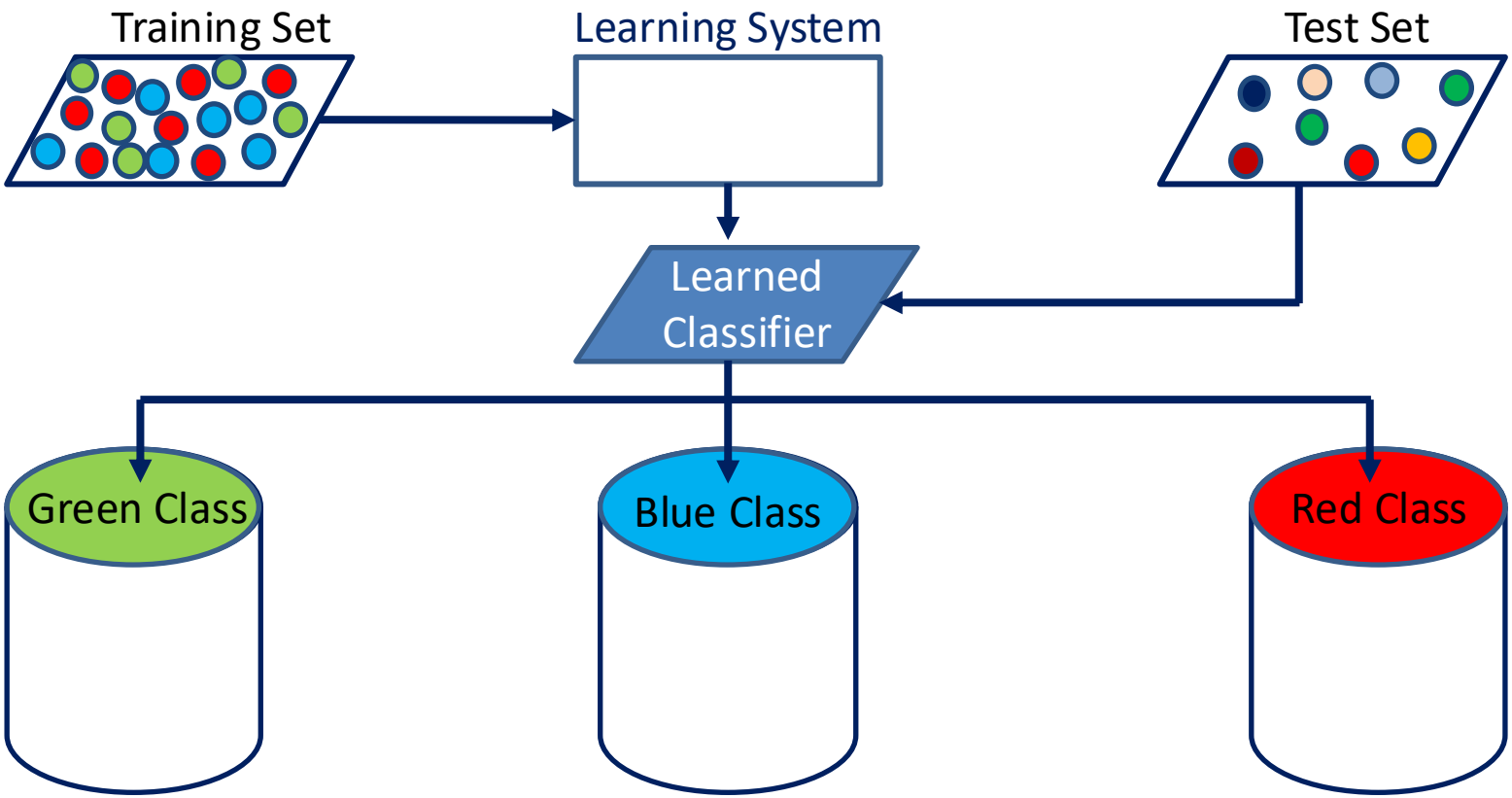
(Definitions)

- Labeled instances (or examples) of the form (x, y)
- $X = (x_1, \ldots, x_{k)}$ is a vector of feature (or attribute) values
- y assumes one of N_c possible different class labels
  - Binary classification = 2 classes
  - Multi-class classification > 2 classes

The classifier builds a **model** (or **hypothesis**) f such that $f(X)$ is the predicted class for an unlabelled instance$(x, ?)$

$$f(X) \rightarrow y$$

Classification Example



Classification Example

Given some **labeled data**, **<u>train</u>** a machine learning **model** that can **<u>classify</u> new unlabeled data** into **one** of several **class labels**

Example: given an image classify whether it is a **0** or a **4**

Labels: 0 0 4 4 0 4

Data:

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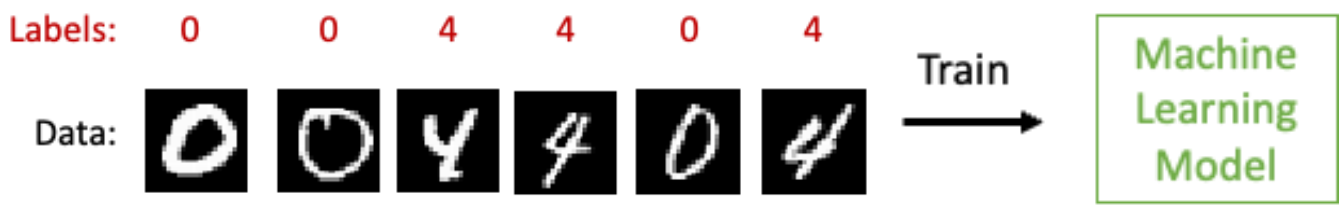

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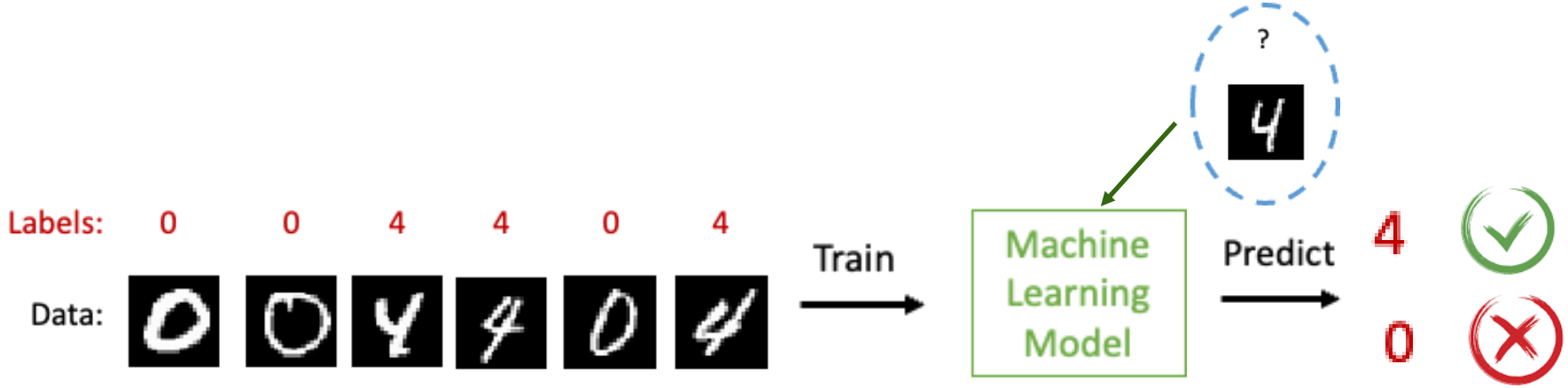
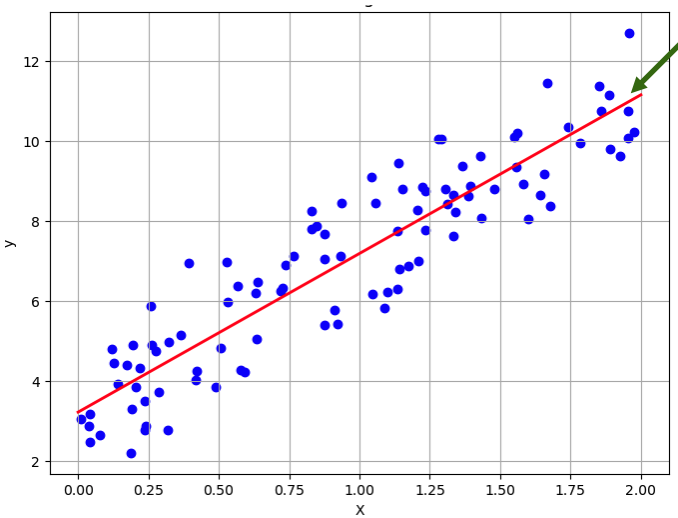


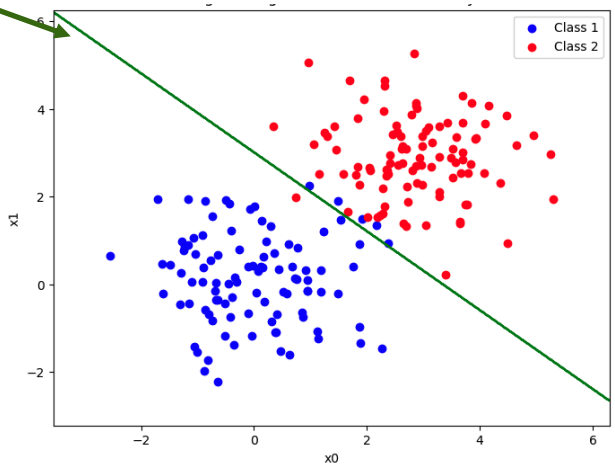
Now we got a concept …

Model

Regression



Classification



Labels: 0 0 4 4 0 4

Data:

Train

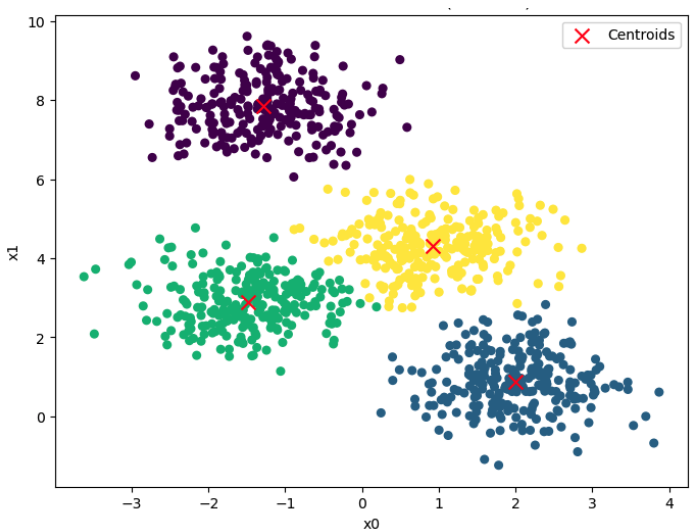
Machine Learning Model

4

0

Clustering

- We still have **input data**, but there is **no ground-truth**
- There is no clear “right” answer, requires interpretation
- The goal is to discover some **hidden structures**, referred to as **clusters**



These are not classes!
Meaning is given to them according to an expert

Cluster != classes

Association Rules Example

Market basket analysis

Transaction 1	bread	marmite	milk	
Transaction 2	bread	butter	banana	
Transaction 3	milk	apple		
Transaction 4	bread	marmite	butter	banana
Transaction 5	bread	marmite	apple	
Transaction 6	bread	marmite	vegemite	apple
Transaction 7	bread	marmite	butter	vegemite
Transaction 8	bread	milk	vegemite	
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Each **transaction** represents something that one customer bought at once

Association Rules Example

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Some possible rules:

$$bread \rightarrow marmite$$

$$bread, butter \rightarrow banana$$

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How do we "quantify" these rules?

Association Rules Example

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$$bread \rightarrow marmite$$

$$bread, butter \rightarrow banana$$

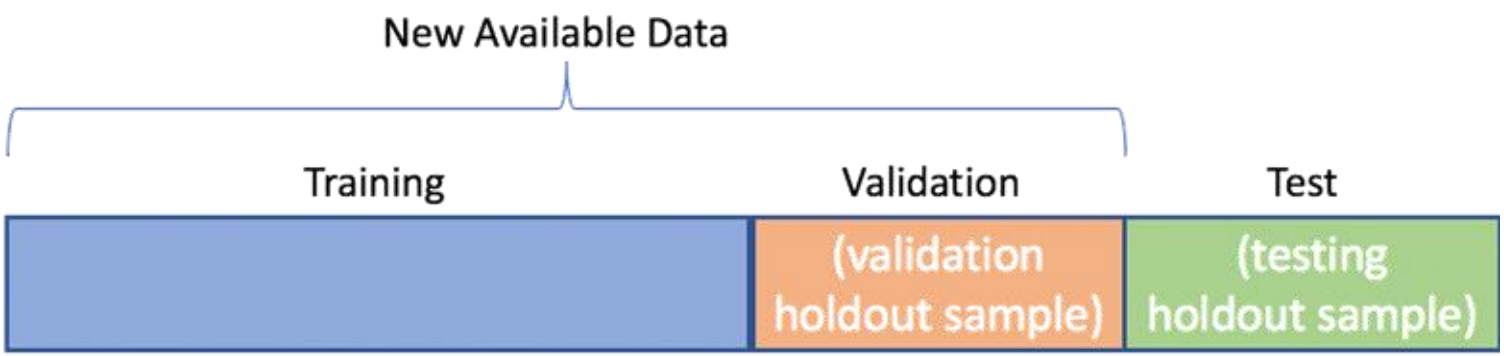
How do we "quantify" these rules?

Co-occurrence/ occurrence

Dataset Splitting

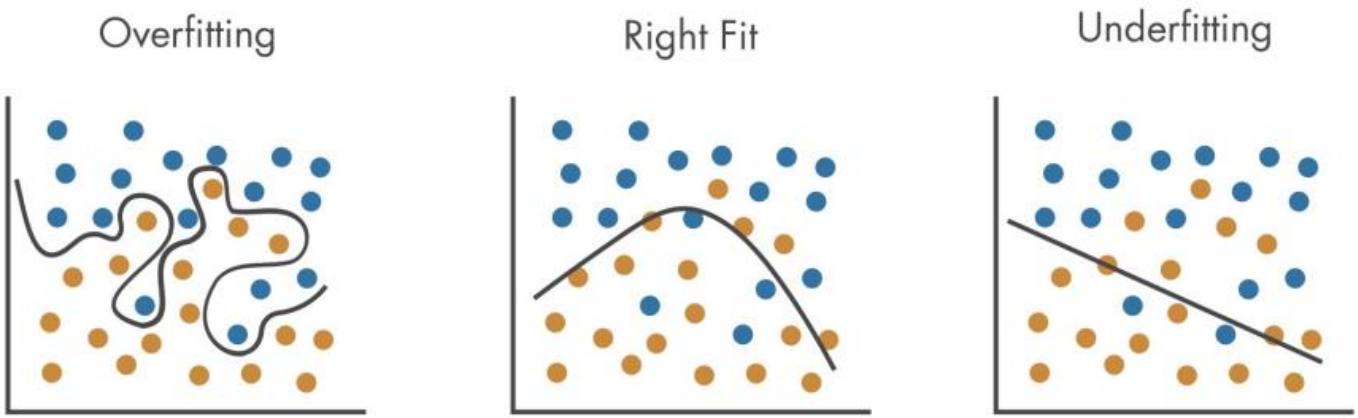
- Training set VS Test set

- Training set, **Validation set**, and Test set
  - Not used for training model and not used for test
  - Unseen data for learned model
  - But used during the training process, e.g., model selection, improve the process of training model

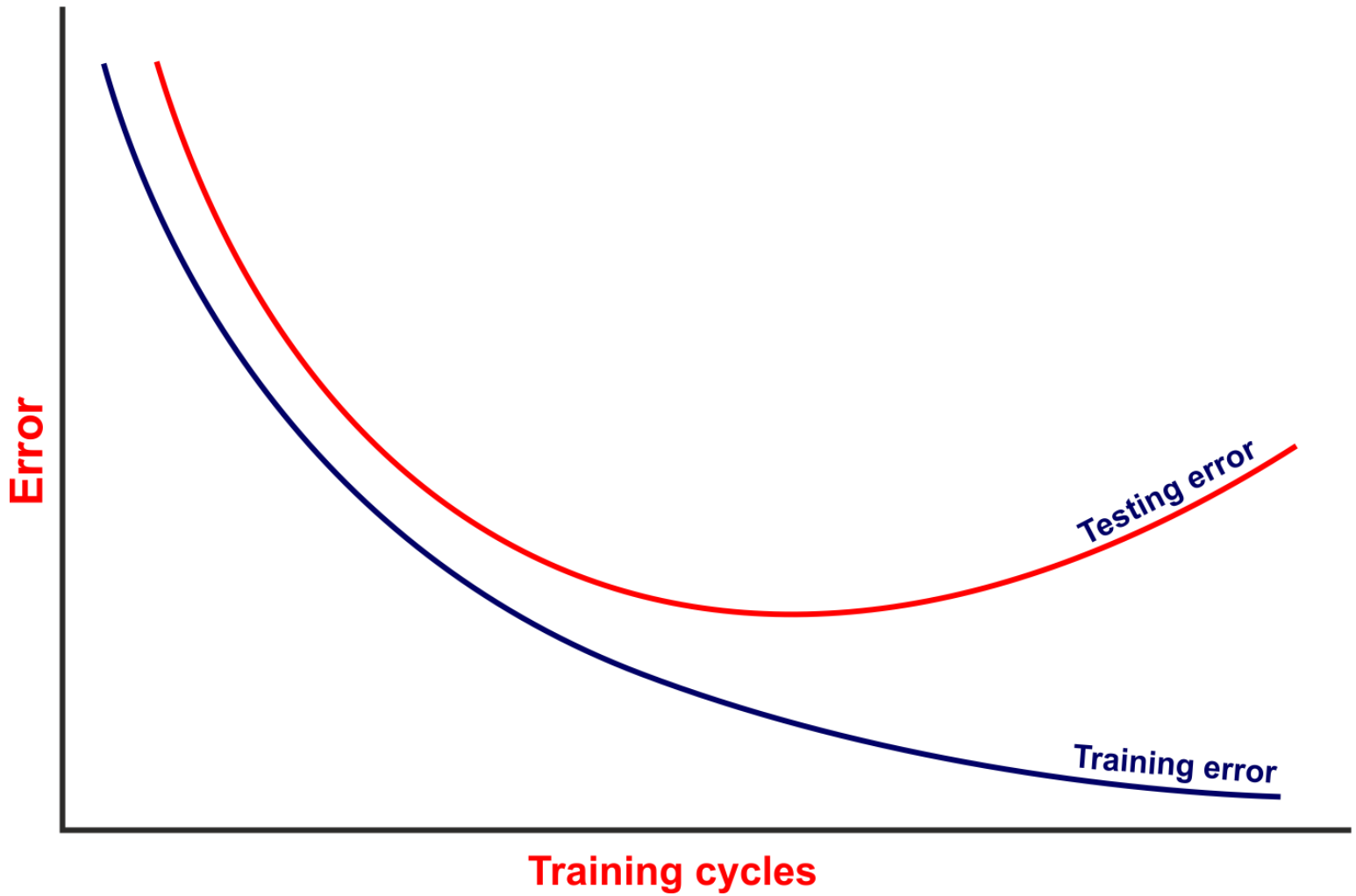


Generalisation

- We learn a classifier/predictor/model from the training data
- But performing well on training data is **NOT** enough!
- Important to evaluate the performance on the **test (unseen) data – generalisation**



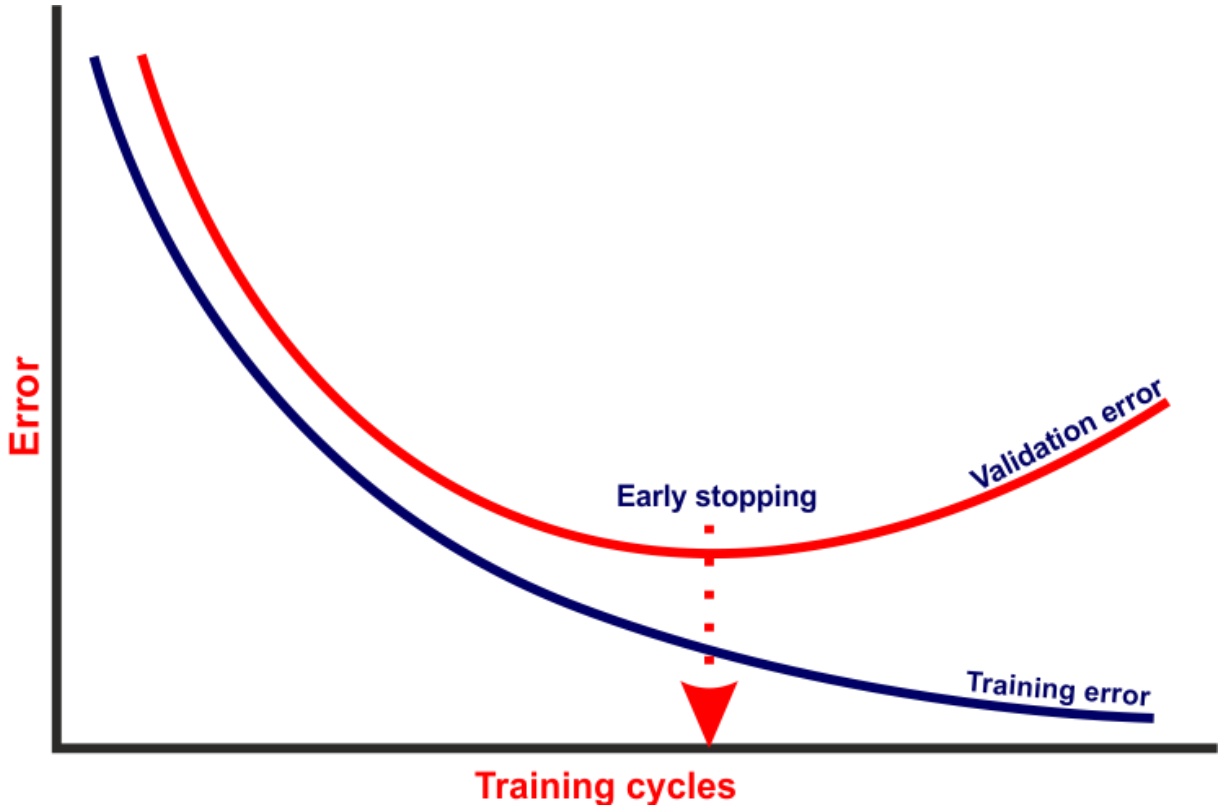
Overfitting



Overfitting

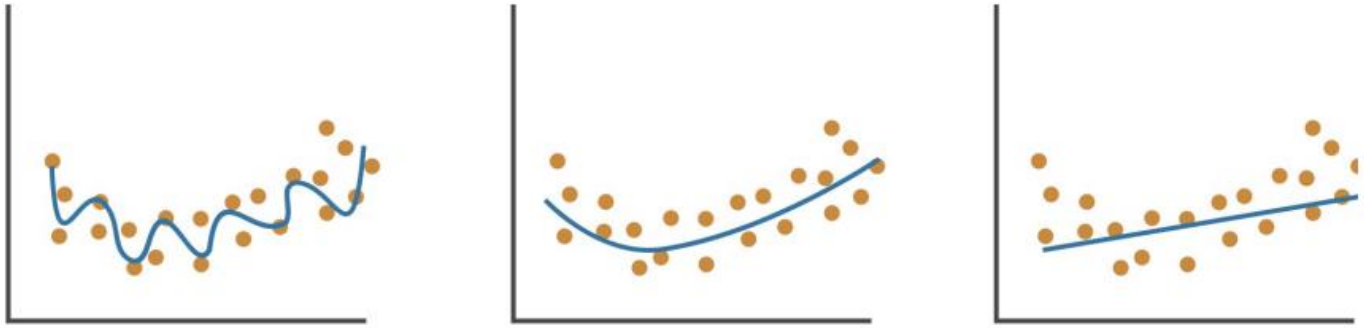
How to handle overfitting:

- Early stopping
- Validation set



Some thoughts

1. See below regression task, underfitting, underfitting, or right fit?



overfitting right fit underfitting

2. Among training data, test data, validation data, which one(s) should not be used to <u>learn</u> models?

Test data and validation data, but validation data can be used to assist the model learning

3. Assume I am working on a research project, can I only include training results for a good report?

No, because test/generalisation is very important for measuring the goodness of algorithms. Only training results do not mean much.

Summary

- Supervised learning and unsupervised learning
- Regression, classification and clustering
- Dataset split: training, test and validation set
- Generalisation and overfitting

Next Lecture:

– K nearest neighbour (KNN)